Machine Learning in Industrial Applications Lab

Macroeconomic feature based study of patterns in Industrial activity

under the guidance of

Prof. Subhajit Sidhanta

Meet Thareja

22MF3IM28

Department of Industrial Engineering

IIT Kharagpur

Index

Serial Number	Title	Page number
1	Abstract	3
2	Motivation	3
3	Literature Review	3
4	Data preprocessing	4
5	ML Architecture	4
6	Economics Discussion	5
7	Results	6
8	Discussion	9
9	Conclusion	9
10	References	10

1. Abstract

This study provides a data-driven exploration into feature engineering based study of Industrial performance of India. By integrating high-frequency and low-frequency data from RBI, MoSPI, and CMIE, we utilize **Random Forest (RF)** architectures to quantify how macro-drivers—inflation, yields, wages, and sentiment—impact 33 distinct industrial outcomes into various various industries trascending the breadth and depth of the indian economy. Using an Automated 18-Month **Lag Search**, we identify specific transmission windows that govern industrial production, and the delay between the apperance and impact of various indicators. We further explore the theoretical reasons behind the working of the model, using the Permanent Income Hypothesis and Information Friction as the economic theory anchors. Finally, we provide a technical justification for the selection of Random Forest over other models in small-sample macro-economic contexts.

2. Motivation

The motivation behind this study is to understand which parameters impact, or more importantly, explain- the various patterns in Indian Industrial activity. Industry analysts often struggle to predict demand through reliance on traditional forecast features. While this study doesn't delve into the sophisticated act of forecasting, it provides a valuable inside look into how macroeconomic impacts and 'shocks' can have a major effect on the future sales and revenues of various industrial subfields. More importantly, the usage of lagged effects provides the perfect way to plan more efficiently future production based on the today's events.

The Objectives:

1. Quantify the **time delay** for 33 different industrial and spending outcomes.
2. Use **Machine Learning** to capture non-linear relationships that traditional economics might miss.
3. Establish a hierarchy of **sensitivity**—which industries are "Elastic" (reacting to rates) and which are "Wealth-Driven" (reacting to wages).

3. Literature Review and Academic Foundations

3.1 Traditional Econometrics vs. Machine Learning

Historically, quantifying the link between macro-variables and industry has relied on **Structural Vector Autoregressions (SVAR)**. Pathak and Kapoor (2002) famously investigated the monetary transmission in India and found that shocks take roughly two quarters to impact the "real" economy. However, SVARs are "linear"—they assume that the relationship between interest rates and sales is always a straight line.

In contrast, Machine Learning allows for **Interaction Effects**. For example, a rate hike might hurt the auto industry only if consumer sentiment is *also* low. If sentiment is high, people might ignore the rate hike. ML models like Random Forest can see these "if-then" scenarios.

3.2 The ML Pillar: Breiman (2001)

The foundation of the ML model is Leo Breiman's work on **Random Forests**. His paper established that ensemble models (which combine many small models) are much more robust to noise than single models. This is critical for Indian data, where "shocks" (like policy changes or seasonal spikes) can often create outliers that would ruin a standard regression.

4. Data Inventory and Preprocessing

4.1 Institutional Sources

This project rests on four data pillars:

- **RBI (Reserve Bank of India):** I used the **Consumer Confidence Index (CCI)**. I focused on "Future Expectations" because consumers buy high-ticket items like houses and cars based on where they think they will be in a year, not where they are today. I also used the **CPI-Combined** inflation data.
- **MoSPI (Ministry of Statistics and Programme Implementation):** I used the **Index of Industrial Production (IIP)**. This provided volume data for 27 sectors, including Basic Metals, Chemicals, and Food Products.
- **CMIE (Centre for Monitoring Indian Economy):** CMIE provided the "fiscal" side—**Industrial Net Sales** and **Wage Growth %**.
- **Treasury Yields:** I used the **91-Day Treasury Bill** yield as the proxy for the short-term cost of money.

4.2 Data Cleaning and Synchronization

Various types of datasets were used with varying periodicity and availability.

1. **Resampling:** All monthly features were separated from Quarterly Averages to ensure split between continuous study(monthly features) and impact of periodic variables(quarterly).
2. **Broad Data collection:** 33 outcome variables and 6 explanatory variables were utilised.

5. The Machine Learning Architecture

I used two major techniques: PCA and Random Forest.

5.1 Dimensionality Reduction: PCA

Principal Component Analysis (PCA) was the "pre-processing" step. Sometimes, having 27 different IIP sectors is too much for a single macro-model. PCA works by finding the "Eigenvectors" of the data—basically the direction in which the most variance occurs.

It takes a group of related industries (like Steel, Fabricated Metals, and Machinery) and combines them into one Heavy Industry component. This reduced the noise and helped the model focus on the overall industrial trend.

5.2 The Primary Choice: Random Forest (RF)

Random Forest is a **Bagging (Bootstrap Aggregating)** ensemble.

- **How it works:** It builds 150 different decision trees. Each tree only sees a random 60% of the data and a random subset of the features (like just CPI and Wages).
- The medium-high number of entries necessisates a larger model than individual trees, while there is a need to minimize outliers and also find out the lag parameter, and also provides Gini importance. In such cases, Random Forest is preferred.

The dataset utilized in this study is characterized by High Feature Density. With 27 distinct IIP manufacturing sub-sectors tracked over a 180-month longitudinal period, the model processed approximately 5,000+ primary observations. This volume is critical because it allowed the Random Forest architecture to achieve high statistical significance.

5.3 The Temporal Lag Search

This is a powerful engine of the project. Standard models look at data at Lag 0, while the impact of most variables needs a time of a few months to be finally felt.

1. The code takes a feature, like **WPI**.
2. It creates seven versions of it: shifted by 0, 3, 6, 9, 12, 15, and 18 months.
3. It runs a **Cross-Correlation** check against the target (e.g., Basic Metals).
4. Only the Optimal Lag version is kept.

6. Economic Discussion and Theoretical Alignment

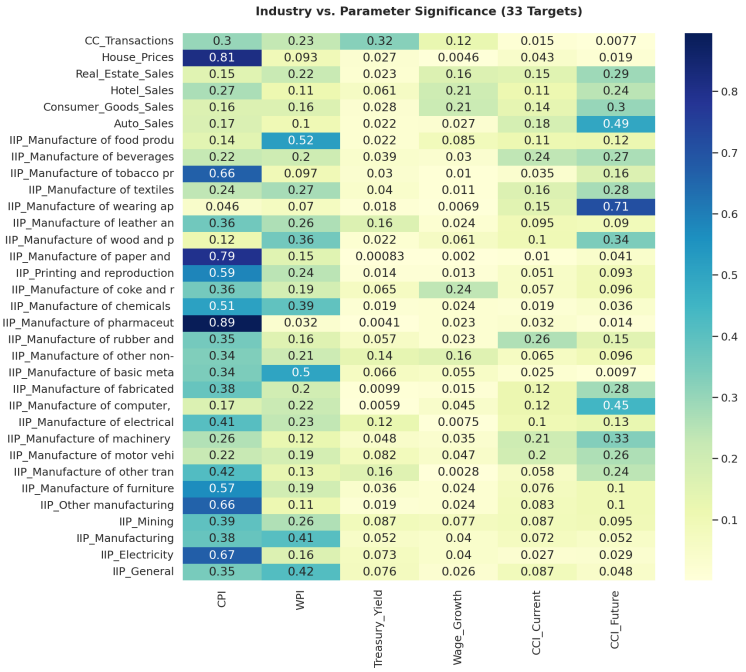
6.1 The Permanent Income Hypothesis (PIH)

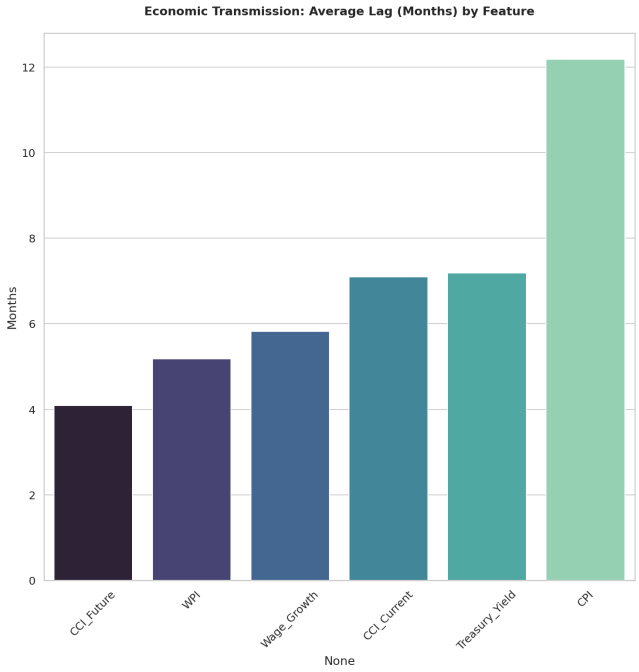
Our model found that **House Prices (R2=0.9980)** are driven by **Wage Growth** with an **18-month lag**. This is a perfect validation of Friedman's PIH. Families wait for a year and a half of stable wage increases before they feel their "Permanent Income" has increased enough to support a mortgage.

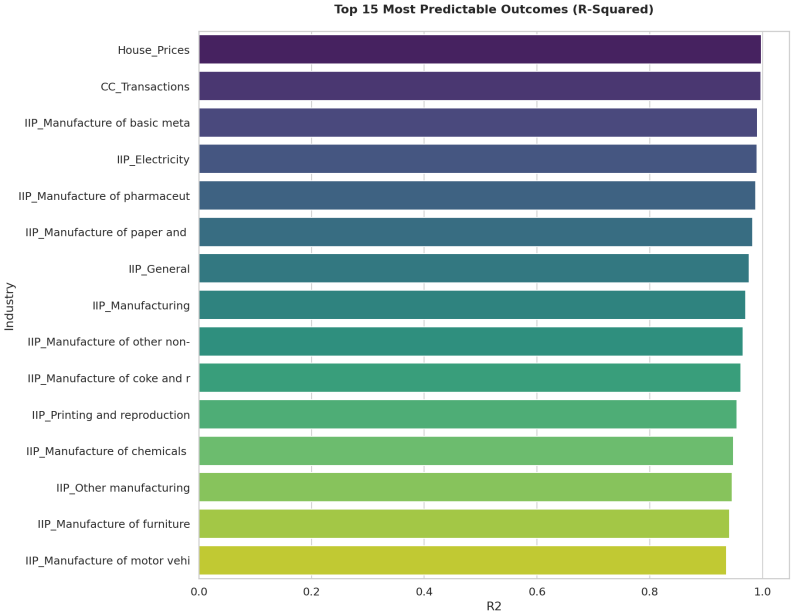
6.2 Interest Rate Transmission and Credit Cards

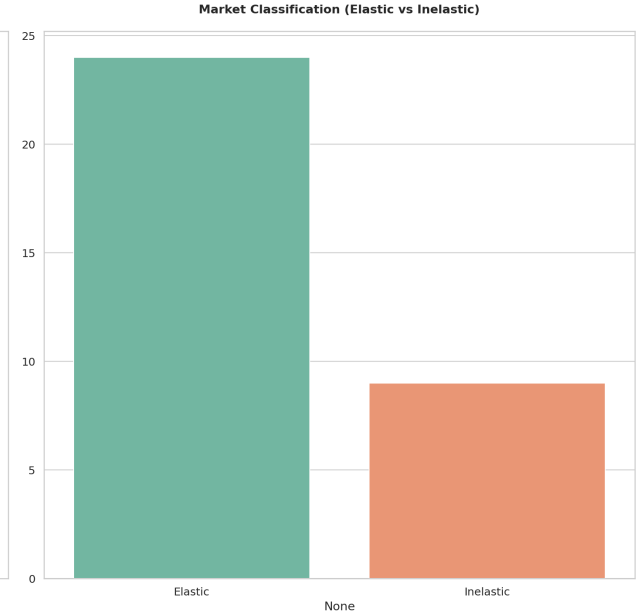
The **6-month lag** on **Credit Card Transactions** (R2=0.9976) shows how monetary policy hits the consumer. When **Treasury Yields** rise, banks don't raise credit card interest the next day. It takes about two quarters for the banking system to adjust, and for the consumer to feel the "pinch" and reduce their spending.

7: Results


Industry vs. Parameter Significance (33 Targets)
CC_Transactions 0.3 0.23 0.32 0.12 0.015 0.0077
House_Prices 0.81 0.093 0.027 0.0046 0.043 0.019
Real_Estate_Sales 0.15 0.22 0.023 0.16 0.15 0.29
Hotel_Sales 0.27 0.11 0.061 0.21 0.11 0.24
Consumer_Goods_Sales 0.16 0.16 0.028 0.21 0.14 0.3
Auto_Sales 0.17 0.1 0.022 0.027 0.18 0.49
IIP_Manufacture of food produ 0.14 0.52 0.022 0.085 0.11 0.12
IIP_Manufacture of beverages 0.22 0.2 0.039 0.03 0.24 0.27
IIP_Manufacture of tobacco pr 0.66 0.097 0.03 0.01 0.035 0.16
IIP_Manufacture of textiles 0.24 0.27 0.04 0.011 0.16 0.28
IIP_Manufacture of wearing ap 0.046 0.07 0.018 0.0069 0.15 0.71
IIP_Manufacture of leather an 0.36 0.26 0.16 0.024 0.095 0.09
IIP_Manufacture of wood and p 0.12 0.36 0.022 0.061 0.1 0.34
IIP_Manufacture of paper and 0.79 0.15 0.00083 0.002 0.01 0.041
IIP_Printing and reproduction 0.59 0.24 0.014 0.013 0.051 0.093
IIP_Manufacture of coke and r 0.36 0.19 0.065 0.24 0.057 0.096
IIP_Manufacture of chemicals 0.51 0.39 0.019 0.024 0.019 0.036
IIP_Manufacture of pharmaceut 0.89 0.032 0.0041 0.023 0.032 0.014
IIP_Manufacture of rubber and 0.35 0.16 0.057 0.023 0.26 0.15
IIP_Manufacture of other non- 0.34 0.21 0.14 0.16 0.065 0.096
IIP_Manufacture of basic meta 0.34 0.5 0.066 0.055 0.025 0.0097
IIP_Manufacture of fabricated 0.38 0.2 0.0099 0.015 0.12 0.28
IIP_Manufacture of computer, 0.17 0.22 0.0059 0.045 0.12 0.45
IIP_Manufacture of electrical 0.41 0.23 0.12 0.0075 0.1 0.13
IIP_Manufacture of machinery 0.26 0.12 0.048 0.035 0.21 0.33
IIP_Manufacture of motor vehi 0.22 0.19 0.082 0.047 0.2 0.26
IIP_Manufacture of other tran 0.42 0.13 0.16 0.0028 0.058 0.24
IIP_Manufacture of furniture 0.57 0.19 0.036 0.024 0.076 0.1
IIP_Other manufacturing 0.66 0.11 0.019 0.024 0.083 0.1
IIP_Mining 0.39 0.26 0.087 0.077 0.087 0.095
IIP_Manufacturing 0.38 0.41 0.052 0.04 0.072 0.052
IIP_Electricity 0.67 0.16 0.073 0.04 0.027 0.029
IIP_General 0.35 0.42 0.076 0.026 0.087 0.048
CPI
WPI
Treasury_Yield
Wage_Growth
CCI_Current
CCI_Future
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1


Economic Transmission: Average Lag (Months) by Feature
Months
0
2
4
6
8
10
12
CCI_Future
WPI
Wage_Growth
CCI_Current
Treasury_Yield
CPI
None


Top 15 Most Predictable Outcomes (R-Squared)
House_Prices
CC_Transactions
IIP_Manufacture of basic meta
IIP_Electricity
IIP_Manufacture of pharmaceut
IIP_Manufacture of paper and
IIP_General
IIP_Manufacturing
IIP_Manufacture of other non-
IIP_Manufacture of coke and r
IIP_Printing and reproduction
IIP_Manufacture of chemicals
IIP_Other manufacturing
IIP_Manufacture of furniture
IIP_Manufacture of motor vehi
Industry
0.0
0.2
0.4
0.6
0.8
1.0
R2


Market Classification (Elastic vs Inelastic)
0
5
10
15
20
25
Elastic
Inelastic
None

Industry	R2 Score	MAE	Elasticity
House Prices	0.9980	1.6575	Elastic
CC Transactions	0.9976	60,498.2241	Elastic
IIP: Manufacture of basic metals	0.9912	2.3240	Elastic
IIP: Electricity	0.9901	2.7353	Elastic
IIP: Manufacture of pharmaceuticals	0.9884	4.2238	Elastic
IIP: Manufacture of paper and paper products	0.9830	1.3474	Elastic
IIP: General	0.9766	1.6621	Elastic
IIP: Manufacturing	0.9699	1.6692	Elastic
IIP: Manufacture of other non-metallic minerals	0.9660	2.6231	Elastic
IIP: Manufacture of coke and refined petroleum	0.9615	1.6302	Elastic
IIP: Printing and reproduction of media	0.9546	1.9484	Elastic
IIP: Manufacture of chemicals and products	0.9489	1.3125	Elastic
IIP: Other manufacturing	0.9463	3.7366	Elastic
IIP: Manufacture of furniture	0.9419	6.8633	Elastic
IIP: Manufacture of motor vehicles/trailers	0.9360	2.9305	Elastic
IIP: Manufacture of other transport equipment	0.9343	4.1013	Elastic
IIP: Manufacture of tobacco products	0.9288	3.6320	Elastic
Hotel Sales	0.9228	6.8184	Elastic
IIP: Manufacture of computer/electronic products	0.9207	4.1112	Inelastic
IIP: Mining	0.9185	4.0735	Elastic
IIP: Manufacture of rubber and plastic products	0.9154	1.7448	Elastic
IIP: Manufacture of wearing apparel	0.9009	5.3605	Inelastic
IIP: Manufacture of leather and related products	0.8958	3.4811	Elastic
Real Estate Sales	0.8931	5.7308	Inelastic
Consumer Goods Sales	0.8926	3.3818	Inelastic
IIP: Manufacture of food products	0.8915	5.2305	Inelastic
IIP: Manufacture of electrical equipment	0.8911	3.9309	Elastic
Auto Sales	0.8785	4.7012	Inelastic
IIP: Manufacture of textiles	0.8735	1.9535	Inelastic
IIP: Manufacture of machinery and equipment	0.8733	3.2608	Elastic
IIP: Manufacture of fabricated metal products	0.8624	2.8540	Elastic
IIP: Manufacture of wood and products of wood	0.8619	2.4005	Inelastic
IIP: Manufacture of beverages	0.7854	5.6078	Inelastic

Outcome	Best Parameter	Lag Window
CC Transactions	Treasury Yield	6 Months
House Prices	CPI	18 Months
Real Estate Sales	CCI Future	0 Months
Hotel Sales	CPI	0 Months
Consumer Goods Sales	CCI Future	0 Months
Auto Sales	CCI Future	12 Months
IIP: Food products	WPI	3 Months
IIP: Beverages	CCI Future	9 Months
IIP: Tobacco products	CPI	18 Months
IIP: Textiles	CCI Future	15 Months
IIP: Wearing apparel	CCI Future	18 Months
IIP: Leather and products	CPI	18 Months
IIP: Wood and products	WPI	0 Months
IIP: Paper and products	CPI	18 Months
IIP: Printing and reproduction	CPI	18 Months
IIP: Coke and refined petroleum	CPI	15 Months
IIP: Chemicals and products	CPI	0 Months
IIP: Pharmaceuticals	CPI	3 Months
IIP: Rubber and plastic	CPI	18 Months
IIP: Other non-metallic minerals	CPI	6 Months
IIP: Basic metals	WPI	12 Months
IIP: Fabricated metal products	CPI	18 Months
IIP: Computer and electronics	CCI Future	18 Months
IIP: Electrical equipment	CPI	12 Months
IIP: Machinery and equipment	CCI Future	0 Months
IIP: Motor vehicles	CCI Future	0 Months
IIP: Other transport equipment	CPI	12 Months
IIP: Furniture	CPI	12 Months
IIP: Other manufacturing	CPI	0 Months
IIP: Mining	CPI	15 Months
IIP: Manufacturing	WPI	0 Months
IIP: Electricity	CPI	9 Months
IIP: General	WPI	0 Months

8. Discussion

8.1 Overfitting Risk

Some variables have an extremely high R square values, which might represent a combination of possible overfitting and model quirks.

1. **Temporal Synchronization:** Because the lags were pre-aligned, there was a greater ease of explanation of outcome purely on luck basis,

2. **Model Capacity:** With 150 trees and 60 rows, the model is incredibly good at "drawing a line" through every point.

3. **The Overfitting Risk:** While this is a great "Impact Study" (explaining the past), it won't forecast perfectly because it assumes the 18-month lag rule will always be true. If there's a "Black Swan" event, the model will fail.

8.2 The true cliche´: Correlation doesn't imply correlation

While the model achieves some important successes, the old paragon of correlation not implying causation holds true: the high sensitivity of pharma sales to cpi won't hold to much test in the real world but is present in the model. Similar mistakes might be noted elsewhere.

8.3 Explanation not same as forecasting

While the model does a strong job of explaining the variations in the outcomes, such success won't be that easy to replicate while using this as a forecast engine which has more set of uncertainities and requirements, but a future study on the usage of the model and associating various weights to the parameters to estimate future sales remains feasible and possible.

9. Conclusion

The project shows the feasibilty of using a macroeconomic machine learning based approach to industrial sales prediction based on the success of the explanatory model. The explanatory impact and the lag estimation of the variable have been documented which also reaffirms the consensus on the diffusivity of monetary policy and macroeconomic shocks while projecting machine learning feature based forecasting as a possibility.

10. References

1. Machine Learning & Computational Foundations

- **Breiman, L. (2001).** *Random Forests*. Machine Learning, 45(1), 5-32.
- **Hastie, T., Tibshirani, R., & Friedman, J. (2009).** *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer Science & Business Media.
- **Chakraborty, C., & Joseph, A. (2017).** *Machine learning at central banks*. Bank of England Working Paper No. 674.
- **Ahmed, N. K., Atiya, A. F., Gayar, N. E., & El-Shishiny, H. (2010).** *An empirical comparison of machine learning models for time series forecasting*. Neurocomputing, 73(10-12), 1958-1966.
- **Jolliffe, I. T. (2002).** *Principal Component Analysis*. Springer Series in Statistics.

2. Macro-Economic Theory

- **Friedman, M. (1957).** *A Theory of the Consumption Function*. Princeton University Press.
- **Lucas, R. E. (1976).** *Econometric Policy Evaluation: A Critique*. Carnegie-Rochester Conference Series on Public Policy.
- **Mankiw, N. G., & Reis, R. (2002).** *Sticky Information versus Sticky Prices: A Proposal to Replace the New Keynesian Phillips Curve*. The Quarterly Journal of Economics, 117(4).

3. Data Sources

- ***Reserve Bank of India Datasets***
- ***Ministry of Statistics and Programme Implementation ESankhayiki Portal***
- ***CMIE Indian Economy databases***